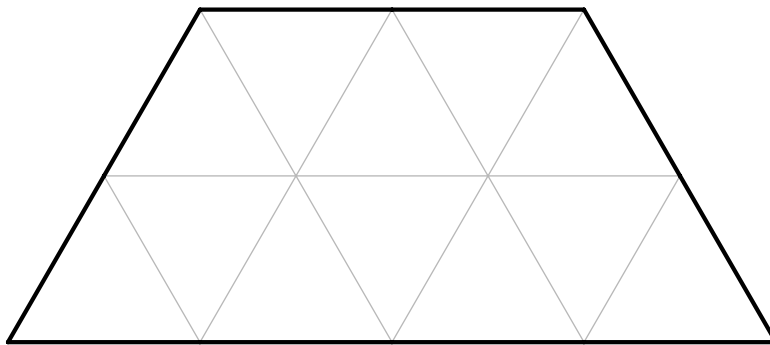
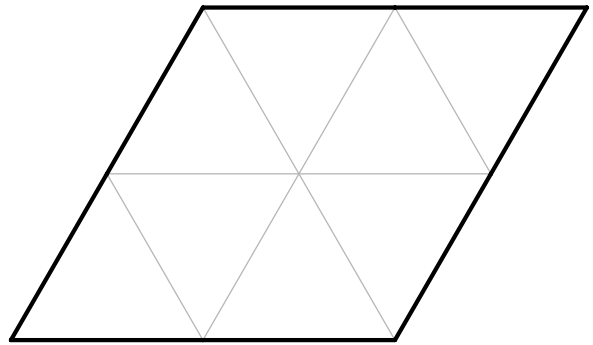
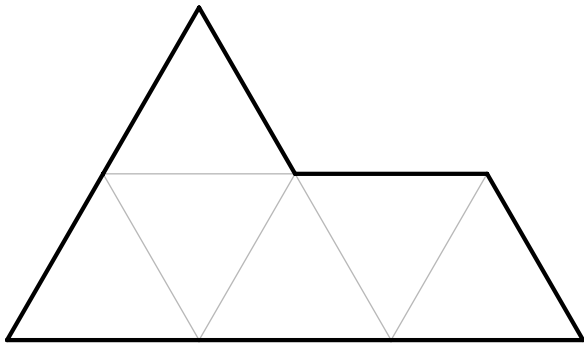
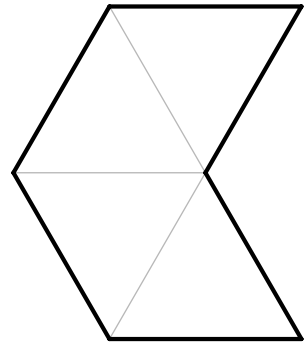
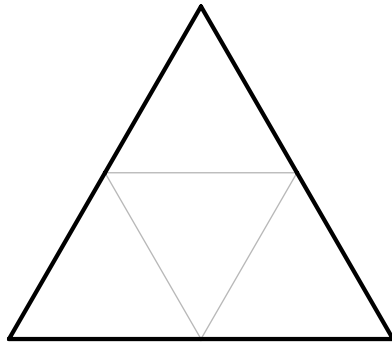
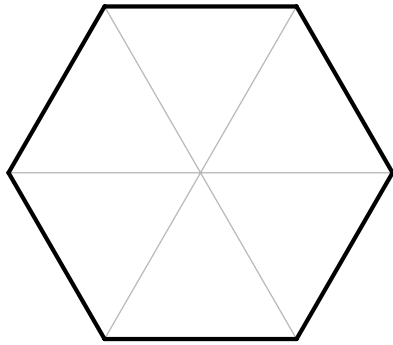
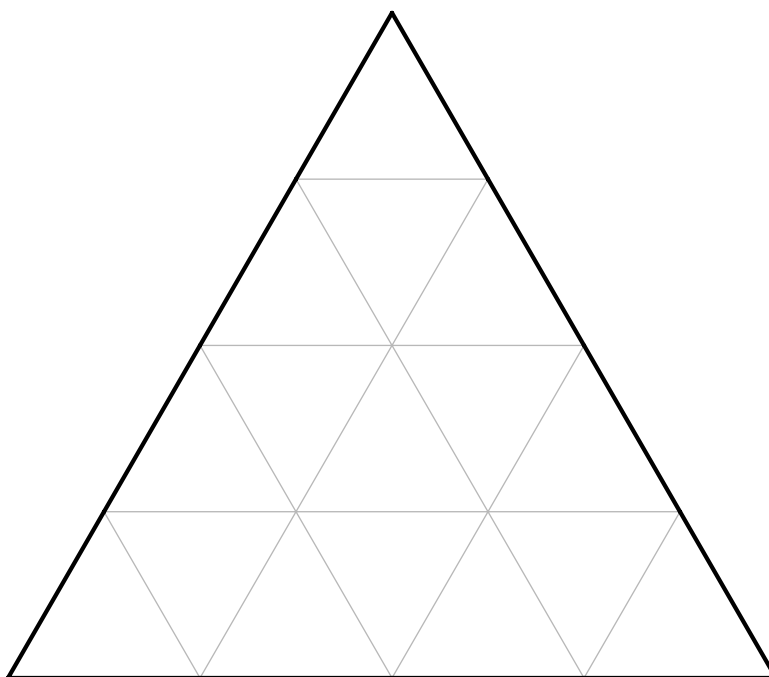
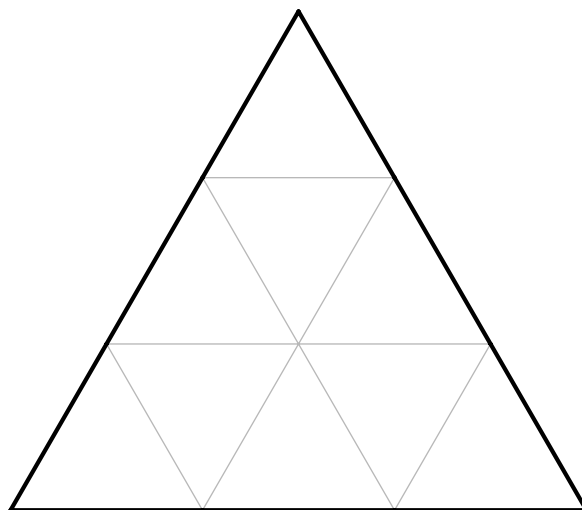
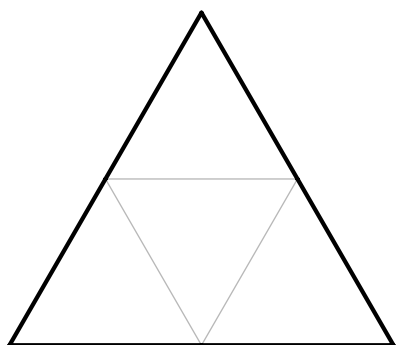


In every problem, blocks go on the lines of the shape. They must stay inside it and must not overlap.

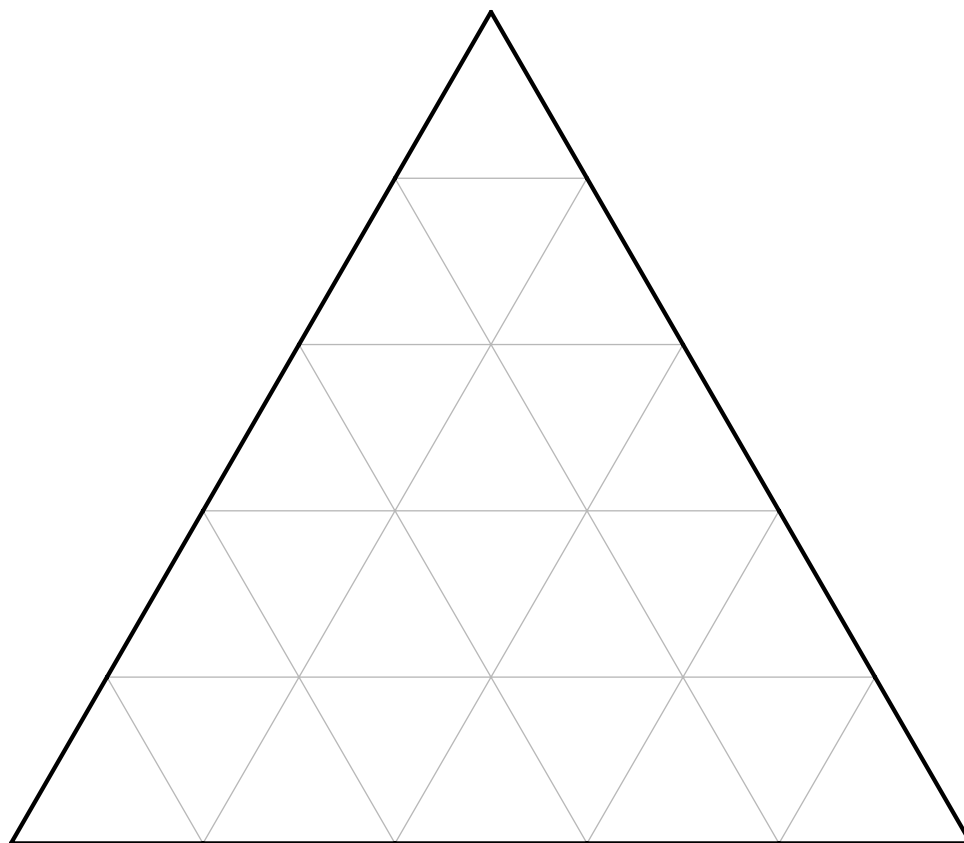
Problem 1: Cover each shape with blue blocks only. Put an X on each shape that cannot be covered.



Problem 2: Cover each triangle with blue blocks and green blocks, using as few green blocks as you can. Write how many green blocks you used.



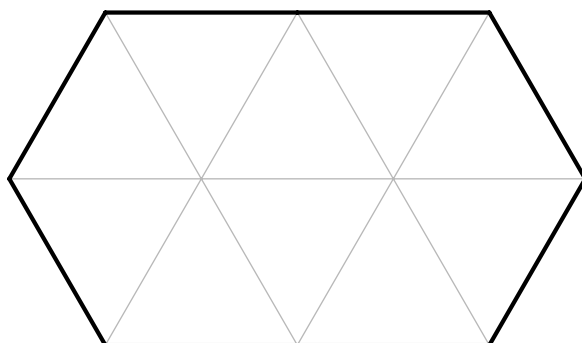
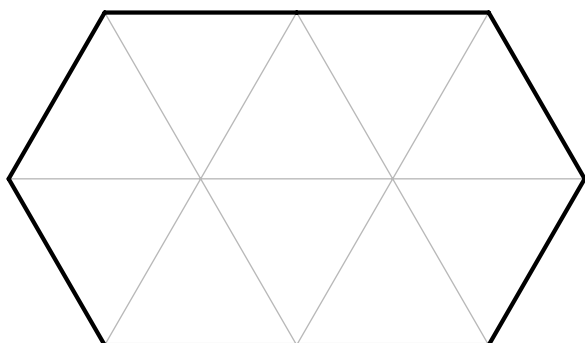
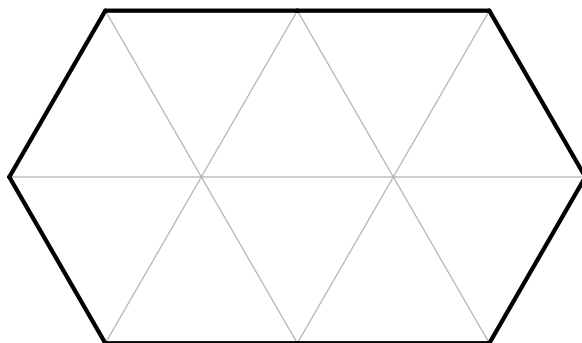
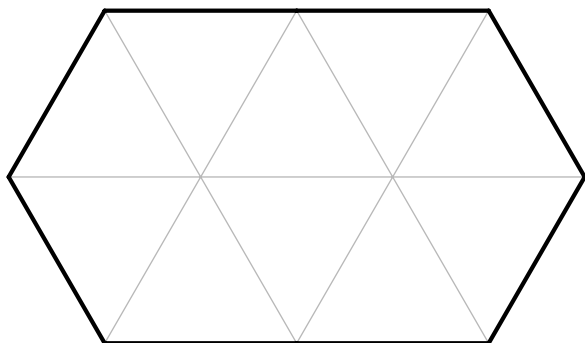
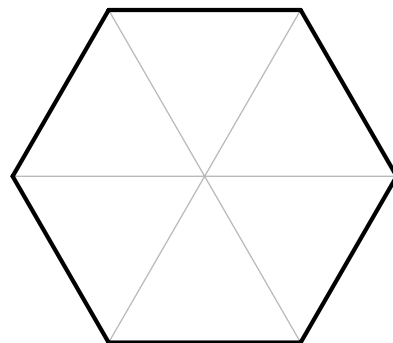
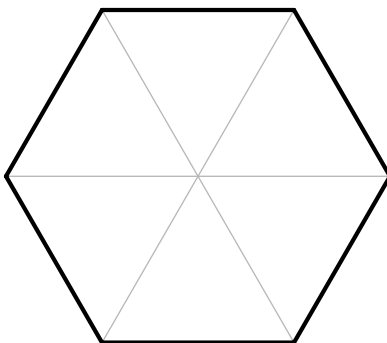
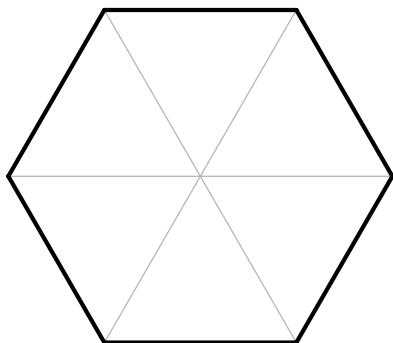
Problem 3: Guess the fewest green blocks you will need for this big triangle. Then cover it with blue blocks and green blocks, using as few green blocks as you can.



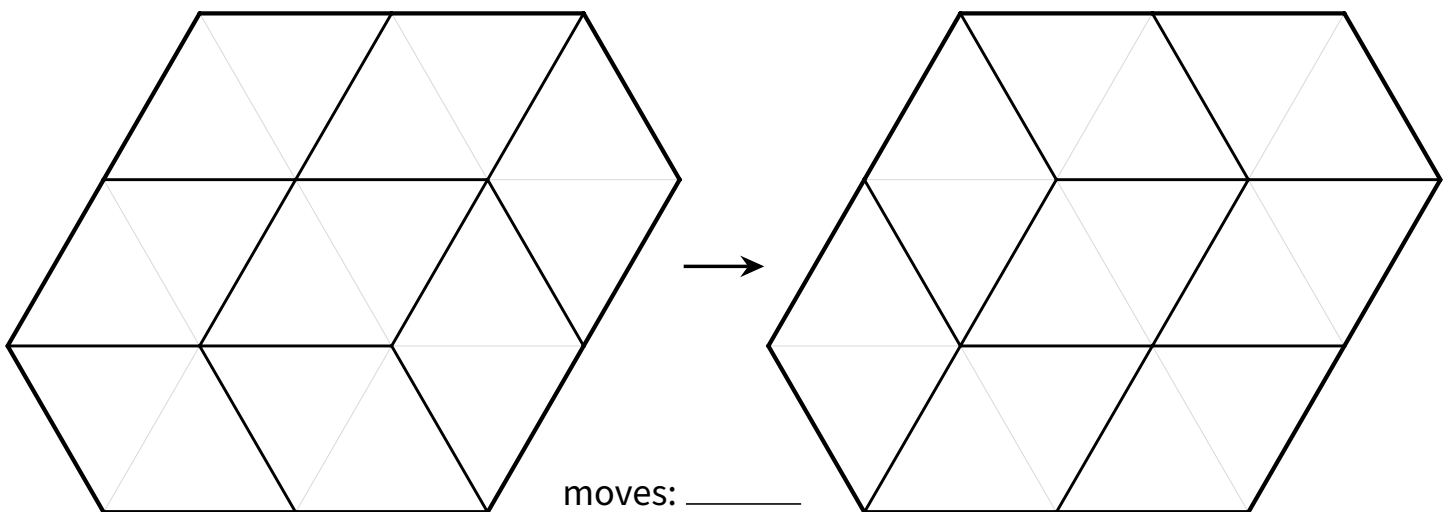
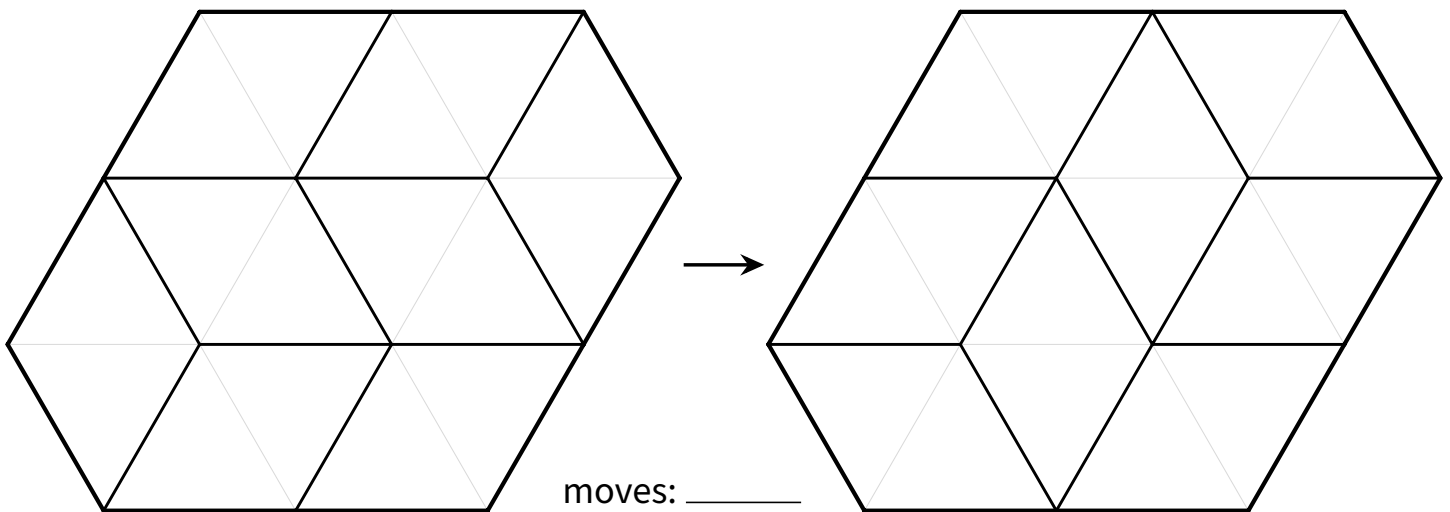
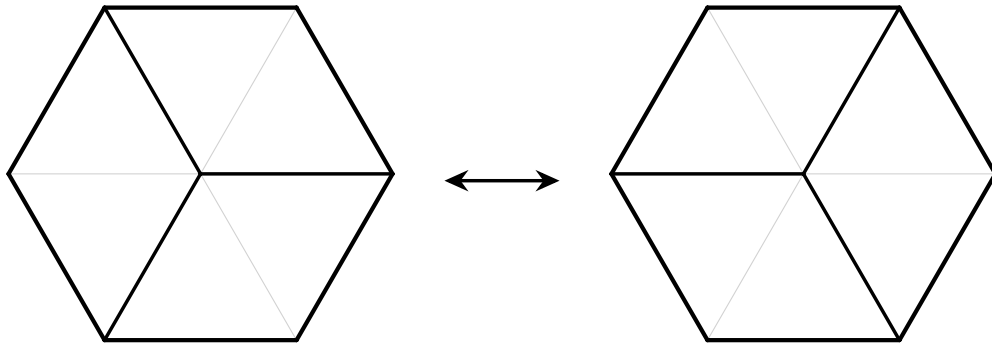
My guess: _____

I used:  _____

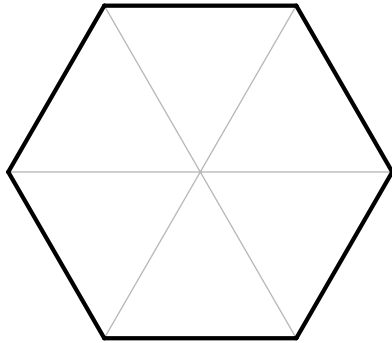
Problem 4: Find every way to cover the hexagon with blue blocks, and every way to cover the long shape. Draw each way in a different copy. Some copies may stay empty.



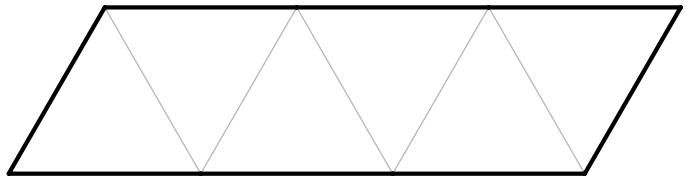
Problem 5: Build the left picture with blue blocks. Change it into the right picture in as few moves as you can. A move turns three blue blocks that make a hexagon, like this:



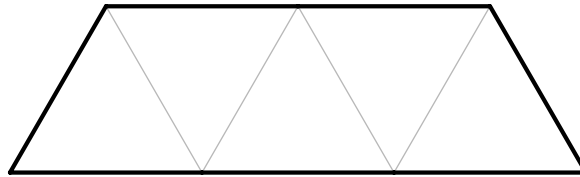
Problem 6: Take turns with a partner putting one green block or one blue block on the shape. The player who puts down the last block wins. For each shape, circle the player who can always win: first or second.



first second



first second



first second